

# House Price Prediction - Summary Report

Internship Project Week 1 | Dataset: Housing.csv | Submitted by: Pratham

## Dataset & Cleaning

The dataset contains 545 housing records and 13 columns. **price** is the target variable. The useful features include area, bedrooms, bathrooms, stories, parking, mainroad, guestroom, basement, hotwaterheating, airconditioning, prefarea, and furnishingstatus. There were no missing values. Duplicate rows were checked and categorical yes/no and furnishing fields were converted into numeric variables using one-hot encoding.

## Model Results

| Model                   | MAE        | RMSE       | R <sup>2</sup> Score |
|-------------------------|------------|------------|----------------------|
| Linear Regression       | ₹970,043   | ₹1,324,507 | 0.653                |
| Random Forest Regressor | ₹1,010,647 | ₹1,391,619 | 0.617                |

**Best model:** Linear Regression. The best model achieved an R<sup>2</sup> score of 0.653, meaning it explains about 65.3% of the variation in house prices. Its average error is around ₹9.70 lakh, so it is useful for rough price estimation but not final valuation alone.

## Insights

- Area was the strongest factor influencing price, followed by bathrooms, air conditioning, parking, and stories.
- The price distribution is right-skewed: most houses are in the lower-to-mid price range, while a few premium houses are much costlier.
- Random Forest did not beat Linear Regression on this split, which suggests that this small dataset has many fairly linear relationships.
- A real estate business should highlight area, bathroom count, air conditioning, parking, preferred area, and furnishing status in listings because these features strongly affect perceived value.

## Business Recommendation

Use this model as an initial price-estimation tool for sellers and buyers. For better accuracy in production, add more location-specific features such as exact locality, distance from transport, property age, amenities nearby, and recent market transaction prices.